

Columbia University - ECBME4090 - BRAIN COMPUTER INTERFACES LAB
Mini Project 1: ERP Decoding and P300-Based Brain–Computer Interface

Student: Alexander Speer

Lab Section: Group 2 (Tuesday Session)

Date of Submission: November 2025

Introduction (4 pts)

Write a short description of the experimental paradigm.

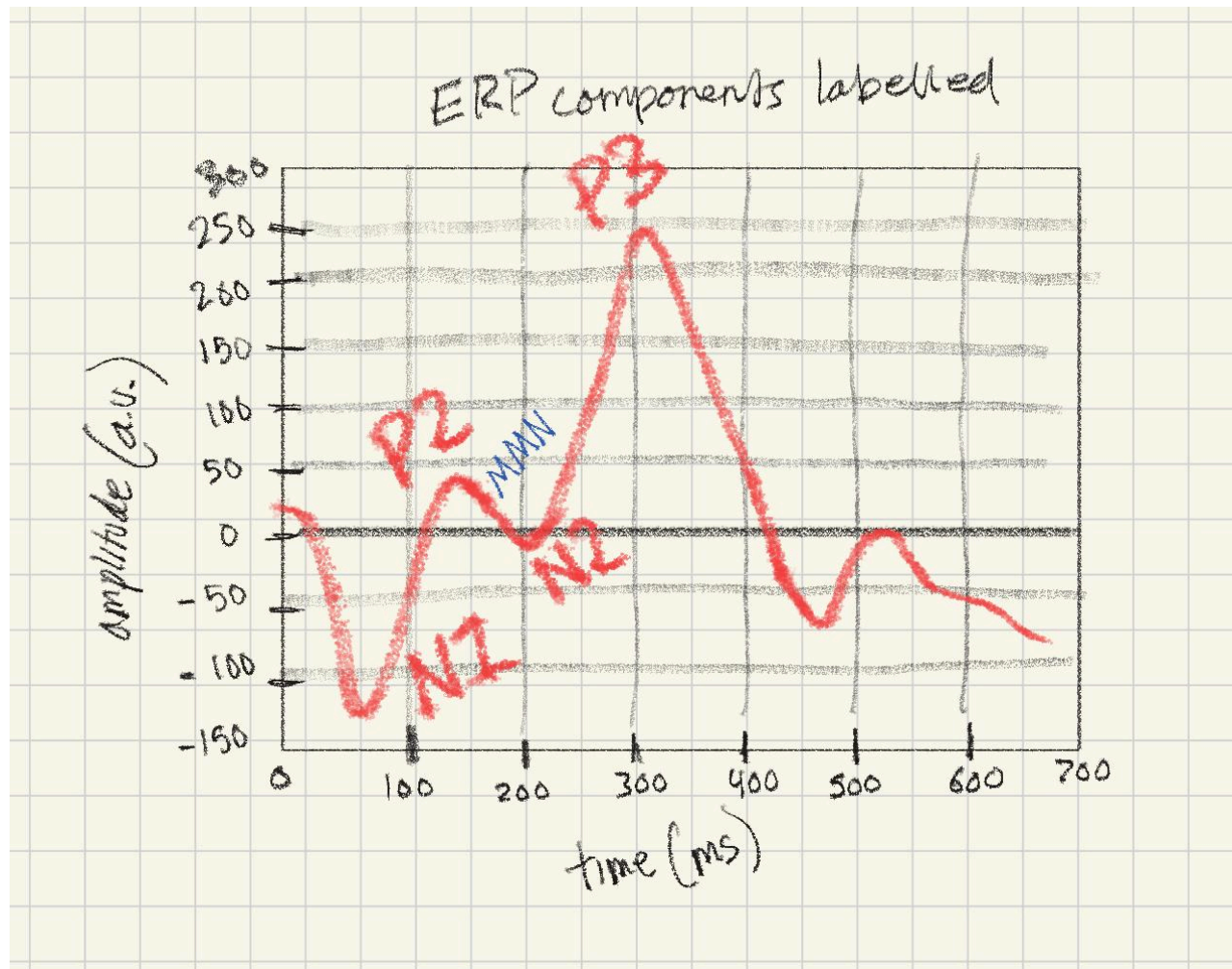
Explain what ERP is and how it is measured. Draw a picture and label the ERP components. Define and explain P300 and mismatch negativity (MMN). What are the differences between the two components?

It is constructive to think of the P300 Speller paradigm as one of the more well-known brain–computer interface (BCI) tasks. It is designed to translate one's neural signals into a means of communication for users (or patients) who find themselves unable to move or to speak. During this particular implementation of the P300 speller experiment, a grid of letters flashes rapidly in a random arrangement of row–column sequences, and then the subject is meant to silently focus on a target letter while counting its flashes. Each time the attended row or column flashes, the hope is that we would be able to discern a distinct positive voltage deflection in the EEG data approximately 300 ms after the onset of our stimulus. This aforementioned positive deflection is what we call the P300 event-related potential (ERP). The goal of the experiment is that if we are able to collect EEG data across multiple electrodes (in this lab: Fz, Cz, P3, Pz, P4, PO7, Oz, PO8), then we will then be able to average the time-locked responses to each flash to extract the ERP, which would then embody the brain's stereotyped electrical reaction to specific sensory or cognitive events.

As I just previously mentioned, an ERP is obtained by way of averaging many trials aligned to stimulus onset, in the hopes that this will cancel the random noise and then highlight consistent neural components such as N100, P200, and P300. Each of those neural components is named after the ms time delay after the onset of the stimulus, meaning that the desired P300 component is a large positive peak around 300 ms, which is typically maximal over the parietal sites, and is thought to reflect the attentional engagement, as well as the brain's evaluation of a meaningful or somewhat unexpected stimulus. In contrast, another term deemed the Mismatch Negativity (MMN), happens to occur earlier (from around 100–250 ms) and is a negative deflection thought to reflect the brain's automatic detection of a deviation in an otherwise repetitive sequence, even without any form of conscious attention. From this, we can intuit that while both are ERPs associated with novelty detection, the P300 depends usually on active attention and task

relevance, whereas the MMN counterpart component reflects a sort of preattentive sensory comparison process.

(Drawing of labeled ERP waveform with N100 (N1), P200 (P1), P300 (P3), and MMN attached below.)



Stimuli and Procedure (2 pts)

Describe your filtering method, sampling frequency, number of experimental trials, how you checked the signal quality, and what the flash and target signals represent.

When our group recorded the EEG signal, we were recording from eight scalp electrodes (named Fz, Cz, P3, Pz, P4, PO7, Oz, PO8), and we were recording at a sampling rate of 256 Hz using the g.USBamp system. Online, through the g.tec software, a 0.5–60 Hz bandpass filter and a 60 Hz notch filter were applied in order to suppress slow drifts and the line noise; whereas in the offline analyses, the channels were detrended and then normalized across all of the trials (when deemed necessary) so that we could improve the comparability. The aforementioned P300 speller stimuli system consisted of a row and column flashes of a 6×6 character matrix that was presented in a pseudorandom order, meaning that each row or column managed to flash $M = 15$ times for the target character, which means that it ended up yielding $12(M)$ individual flash events per character.

In terms of the signal quality, our group went about verifying this in two main ways. First, our group used the Simulink scope was observed during recording to ensure there was some clear alpha activity (around 8–12 Hz) in the occipital electrodes (such as the Oz and the PO channels) when the subject closed their eyes, we could verify such alpha activity through counting these rhythmic oscillations and confirming that there was proper electrode contact and ensuing clean signal acquisition. Second, we also tested the signal quality with some blink and muscle artifact verification, which was performed by having the subject blink and clench their jaw. As we have seen in one of the first labs, these blinks appeared as large frontal deflections, and the jaw movements produced some bursts of high-frequency activity, which would hopefully confirm that all electrodes and amplifiers were in fact functioning properly.

As far as describing what the flash and target signals meant, the “flash” signal encoded which of the rows or columns was illuminated at each time point (1–6 = rows, 7–12 = columns, 0 = no flash), while the “target” signal actually indicated whether that flash corresponded to the attended letter (1 for the target, and 0 for the non-target), serving as our sort of ground-truth label for this subsequent ERP decoding. During our group's data collection, we were instructed to spell specific words in order to generate training and testing datasets. For the train1 data, the target word was “inductor” (with the classifier output being: “otd6c2ur”), followed by a test word “inches” (with classifier output: “inchks”). For train2 interaction the following week, the training word was “flowchartings” (with classifier output: “xbi2ozm3zknsn”), and the testing word was “bcilabrocks2” (classifier output: “boilabrocqy8”).

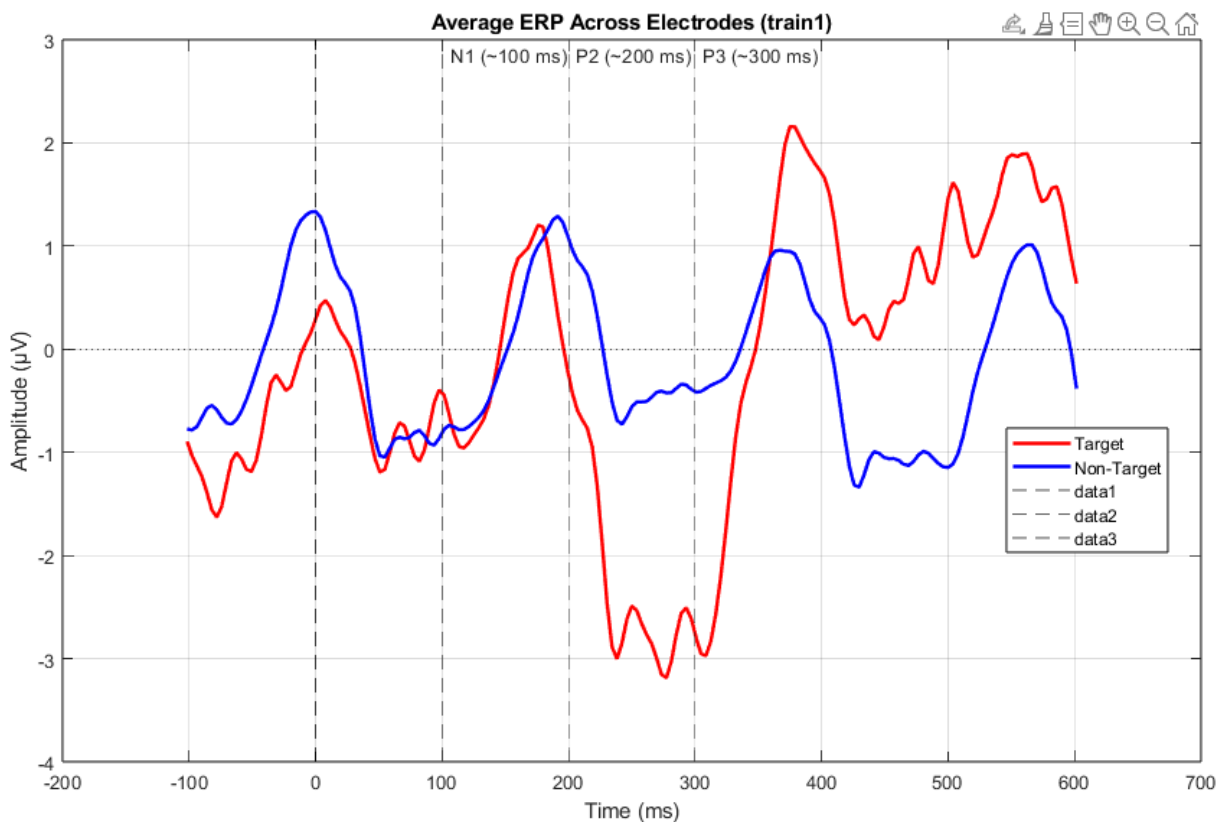
Results

In this section, we will be training and testing linear classifiers to compare model performance in different scenarios. You can use any classifier you like, but $Mdl = \text{fitlinear}(X,Y)$ and $Y_pred = \text{predict}(Mdl,X)$ is suggested for simplicity. Furthermore, you are not required to optimize model hyperparameters (e.g., regularization strength, model type, etc.); however, points will be deducted if none of your models perform above chance.

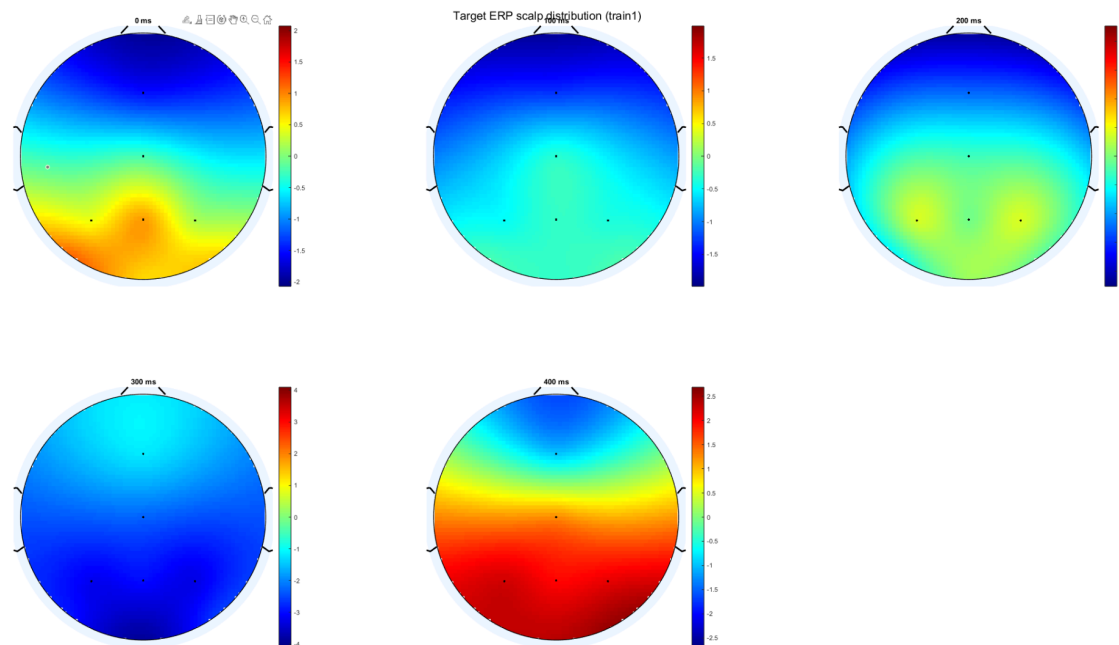
Part 1. Within-Subject, Cross-Session Analysis

ERP Response to Target and Non-Target Flashes

First, using the train1.mat data collected by your group, calculate and plot the average ERP, averaged over electrodes, for target and non-target flashes from 100 ms before to 600 ms after each flash (same as HW). Mark the ERP components visible in the average ERP plots.

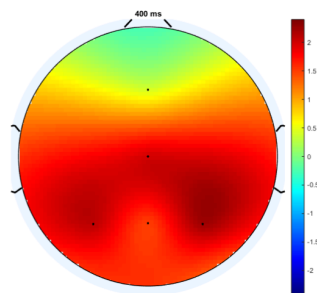
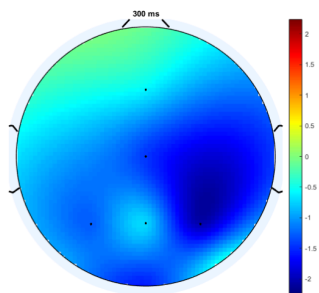
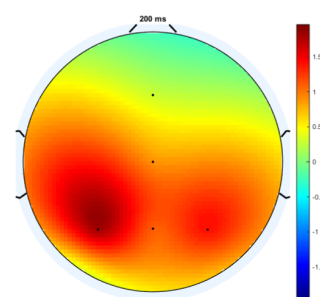
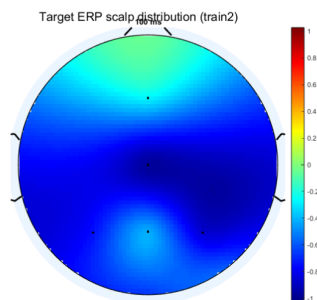
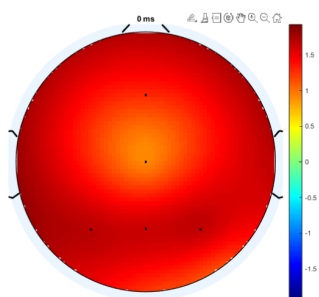
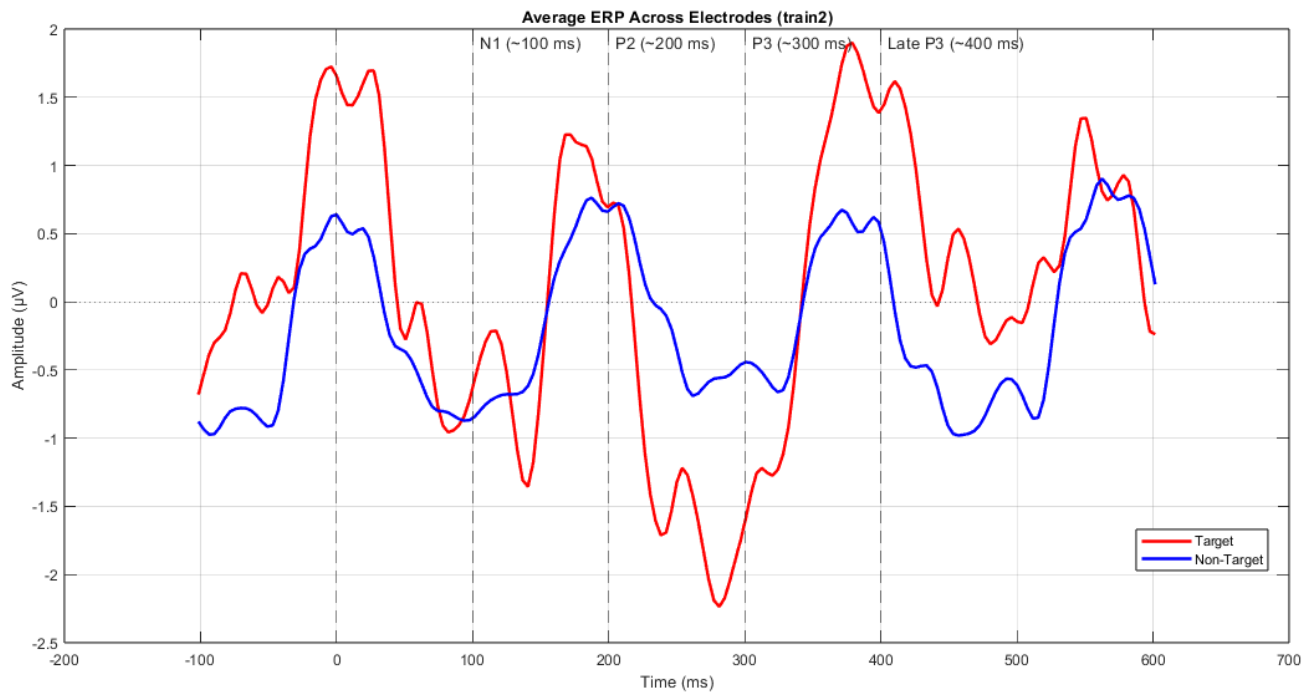


Plot the scalp map of the signal at 0, 100, 200, and 300 ms after the onset of the flash. Use `topoplot.m` from EEGLAB and `BCI.locs` on Courseworks.



For these topoplots, I decided that an additional scalp map at 400 ms should be included because our subjects' P300 component constantly appeared to peak closer to 400 ms in the averaged ERP than the 300-ms component, making this time point seemingly more representative of the maximal positive response.

Produce these plots (average ERPs, marked components, topoplots) again, this time using the *train2.mat* data collected by your group. Do these plots differ from the *train1.mat* plots? If so, how do they differ and why do you think they differ in that way? If not, what makes them so consistent?



From my observations, I can say that the plots from the train2 mat file actually do differ slightly from those in train1. In both sessions, it appears that the P300 peak is slightly delayed, since it seems to be reaching its maximum around 380 ms (which is essentially why I decided that the 400 ms scalp map should be included, since it seems to capture all of the most intense of the parietal–occipital lobe activity). For the train2 data, it appears that earlier responses (around 200 ms) exhibit greater activation in occipital regions, which could suggest something like more robust visual processing. I also think it is worth noting that the 0 ms spike in train2 is unusually large, producing a highly saturated initial topoplot, possibly due to a stronger visual onset response or transient artifact. In contrast, train1’s ERPs are smaller in amplitude and less spatially intense, which may reflect minor variations in attention, arousal, or electrode impedance between sessions.

Training and Testing Within-Subject Decoding Models

Organize each of your datasets (train1/2, test1/2) into a data matrix, in which the rows are trials (individual flash events) and the columns are the time-domain features of the ERP from all channels. Each trial should capture approximately 100 ms before to 600 ms after the flash. How many features are there? Write an equation for the number of features based on the start time relative to the flash (t_{min}), the end time relative to the flash (t_{max}), the sampling rate (fs), and the number of channels (N_{ch}).

Each trial (or distinct flash event) is going to include all of the samples from all of the channels between the t_{min} and t_{max} that are relative to the stimulus onset.

Equation:

$$N_{features} = N_{ch} \times (\text{round}(|t_{min}| \times fs) + \text{round}(t_{max} \times fs) + 1)$$

To elaborate on the variables:

- N_{ch} = number of EEG channels
- t_{min} = is the start time relative to the flash (e.g., -0.1 s)
- t_{max} = the end time relative to the flash (e.g., 0.6 s)
- fs = the sampling rate (Hz)

Each trial is going to have 1448 features (made up of 181 samples $\times$ 8 channels)

Train three models: one using just the train1 data (Model 1), one using just the train2 data (Model 2), and one using both train1 and train2 data (Model 1+2). Test each of these models on just the test1 data, just the test2 data, and both the test1 and test2 data together. Fill in the table below with the accuracy of each model on each test dataset.

Accuracies	Test Set 1	Test Set 2	Test Set 1 + 2
Model 1	0.6972	0.6384	0.6580
Model 2	0.6583	0.6435	0.6485
Model 1 + 2	0.6889	0.6625	0.6713

Based on your knowledge of the P300 and the average ERPs you observed before, select a new time range captured by each trial. What range did you choose, and why? Describe how your results changed after changing the time range. Continue to use your time range for the rest of this section.

When it came to selecting for the new revised analysis window, I decided that it would be best to select a time range from around -0.1 to 0.8 seconds relative to our stimulus onset. This adjustment was based primarily on the observation that the participant's P300 component was always consistently appearing later than the canonical 300 ms latency that we would expect, with the maximal positivity emerging around 380–400 ms. It logically follows then that extending the post-stimulus interval to 800 ms would help to ensure the full capture of the late P300 response while also preserving the pre-stimulus baseline period from the original window we had. I believe that this choice balances the early sensory and also the attentional features (e.g., N1, P2) with all of the later cognitive evaluation that we can expect to be reflected in our delayed P300.

After applying the new time, I saw that the model showed a modest overall improvement in the model's performance, which was particularly evident for the combined model trained on both sessions. Model 1 + 2 improved from 0.6713 to 0.7077 on the combined test set, which could perhaps indicate that by extending the time window, it helped to capture some of those additional discriminative features from our participants' delayed P300 response. Moreover, you could see that the longer window also incorporated more of the late positive activity that could have previously been truncated, while also managing to maintain some comparable accuracy for the single-session models (Model 1: 0.6580 to 0.6506; Model 2: 0.6485 to 0.6920).

(Accuracy table for revised analysis window on next page)

Accuracies	Test Set 1	Test Set 2	Test Set 1 + 2
Model 1	0.68148	0.63519	0.65062
Model 2	0.68704	0.69444	0.69198
Model 1 + 2	0.72222	0.70046	0.70772

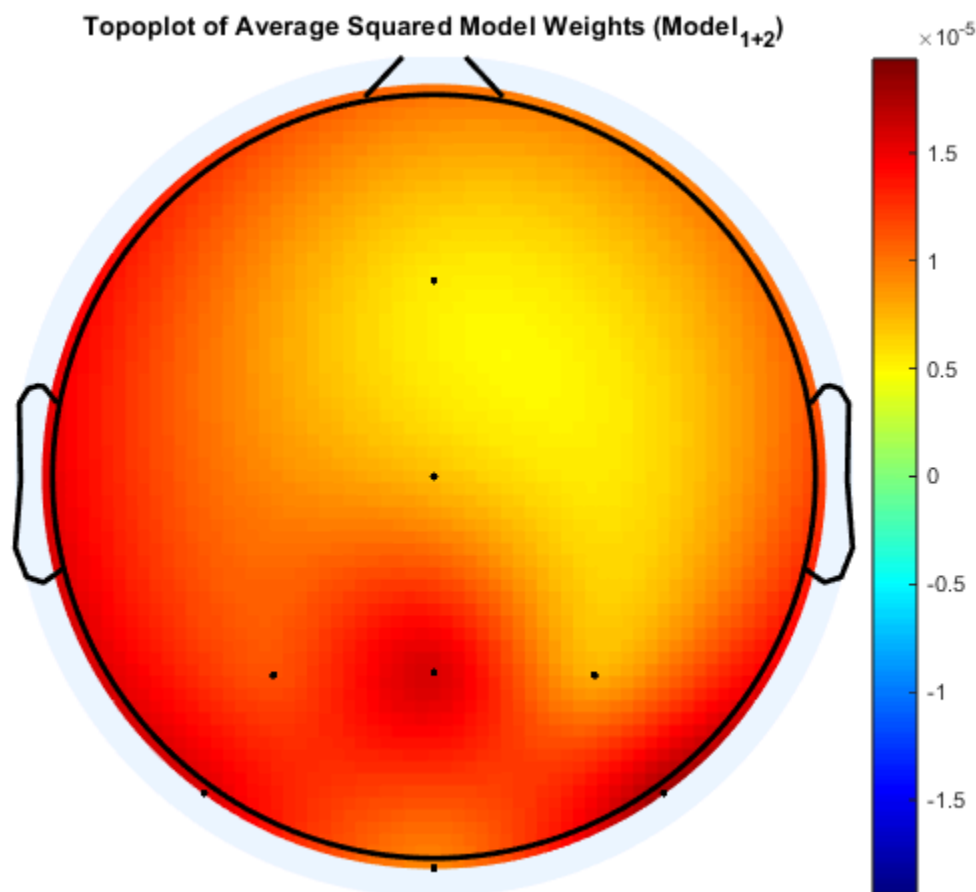
Which (model, test set) combination produced the highest accuracy? Why do you think that is? When we evaluate a model trained using data recorded from one session on test data that was recorded from a different session, we call this generalizing across sessions. How well do your models generalize across sessions? Which model generalizes better? Why do you think that is? When we build models using data from multiple sessions, we call them cross-session models. How does your cross-session model accuracy compare to your within-session model accuracies?

When looking at the table, we can see that the highest accuracy was achieved by Model 1+2 when tested on the combined Test Set 1+2 (70.77%). To me, this result makes sense, as the model was actually trained on data from both of our training sessions, which would naturally give it greater exposure to all of the inter-session variability that we would expect, such as small differences in the electrode impedance, the level of attentiveness, and what background noise was included. I would say that all of these factors would allow this particular model to generalize more effectively when it comes to being evaluated on the merged test data.

It seems that when comparing all of the cross-session generalization, it seems that models trained on a single session (such as Model 1 or Model 2) always performed slightly worse when tested on the opposite session (about 63–69%), which could be indicative of some sort of moderate generalization but also some sensitivity to the session-specific factors. Looking at the accuracies between the two, Model 2 manages to generalize marginally better, likely because the second recording contained cleaner signals, a longer training word, and stronger P300 overall amplitudes.

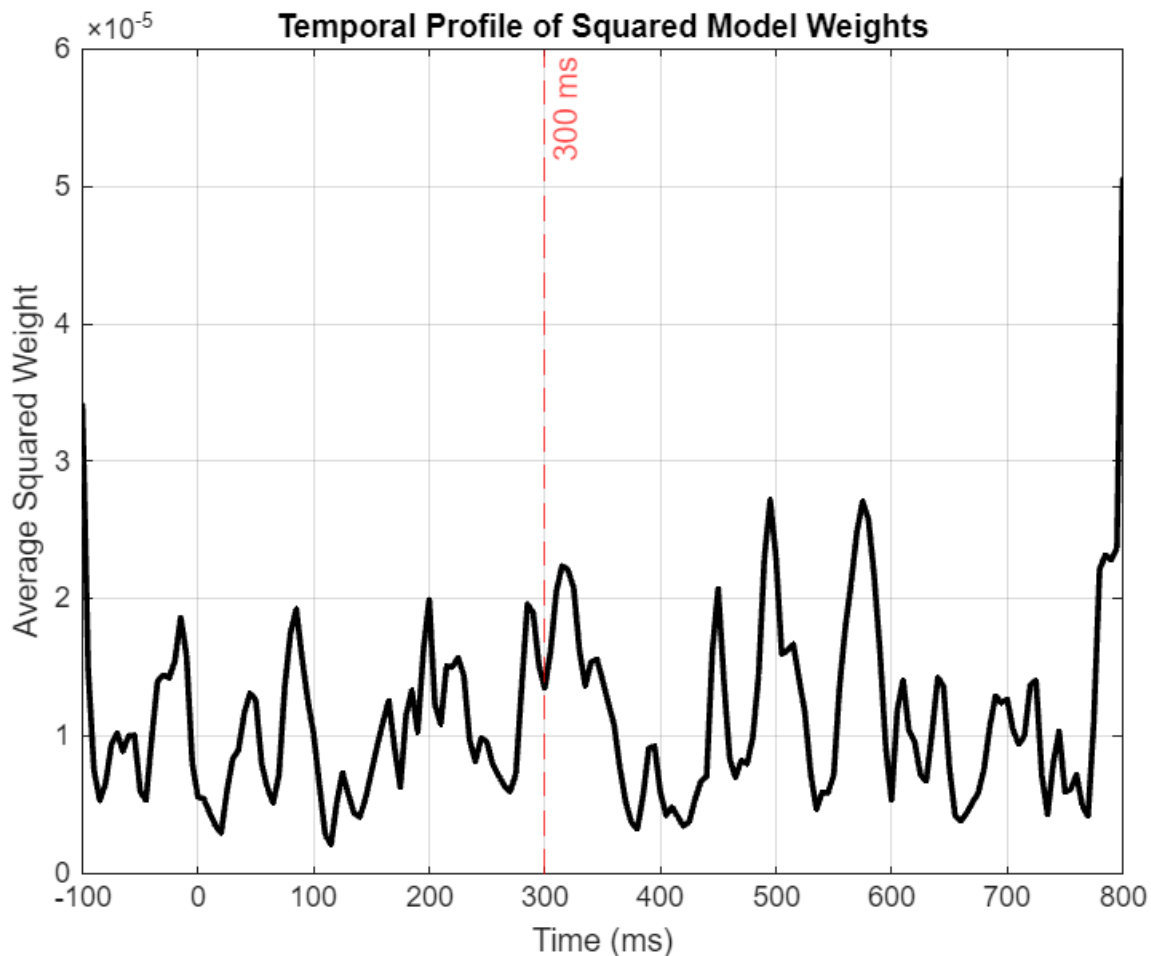
Analyze Within-Subject Decoding Models

Use `Mdl.Beta` to get the weights of your best model. Reshape the weights to a matrix of shape $N_{ch} \times T$, where T is the number of time samples. Square the weights to get the power in the weights. This allows you to average over channels or over time without worrying about whether the weights are positive or negative. Average the squared weights over time, and create a topoplot of the average squared weights at each electrode. How does this compare to the topoplot of the average ERP 300 ms after flash onset? Average the squared weights over electrodes, and plot the average squared weights over time. How does this compare with the corresponding time window of the average ERP plots?



Here we can see the topoplot with the average squared model weights that reveals how the strongest classifier weights were concentrated around and over the parietal–occipital electrodes, with gradually decreasing power as you move towards more of the frontal sites. While I can say that this distribution does not align perfectly with what we saw from the scalp map of the average ERP at 300 ms, I believe that it closely resembles the activity pattern that we see at the 400 ms topoplot, which is the time point where our group 2 participant's P300 component

happened to reach its maximum amplitude. I would also be inclined to say that this similarity offers support to the interpretation that the classifier is most sensitive to spatial patterns associated with the late P300 positivity rather than the actual canonical 300 ms response. To this end, the slightly higher frontal activity that we see in the weight map when compared to the 400 ms ERP could possibly reflect the model actually capturing additional discriminative information from the earlier components, such as the P2 or the sustained positive complex that follows the P300.



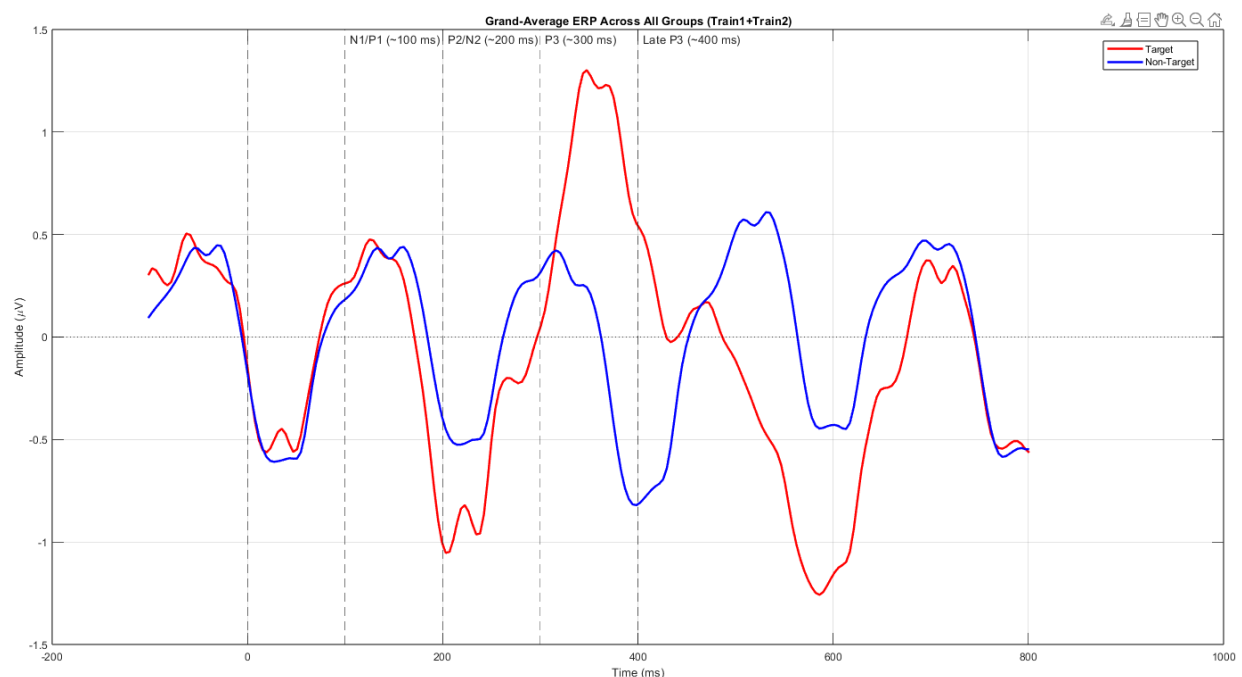
In terms of the temporal profile of the squared model weights, we can see that it shows multiple regions that feature some sort of elevated weight power, with some noticeable activity around 300–350 ms, which would be consistent with the expected latency of the P300, followed by a few strong additional peaks at ~500 ms and ~600 ms. After some thought, I believe that these later increases likely correspond to some array of extended or secondary positive deflections that can be observed in the participant's ERPs, which might reflect continued cognitive evaluation or sustained attention processes rather than a single, sharply defined peak. Additionally, I believe that the fluctuations at the beginning and at the end of the plot would be characteristic of noise

amplification at the temporal boundaries, where we may expect that fewer reliable features are going to end up contributing to the classifier. Overall, I believe that I can say that both the spatial and temporal profiles of the model weights indicate that the classifier primarily relied on the parietal and the occipital lobe activity that occurred between the 300–600 ms mark.

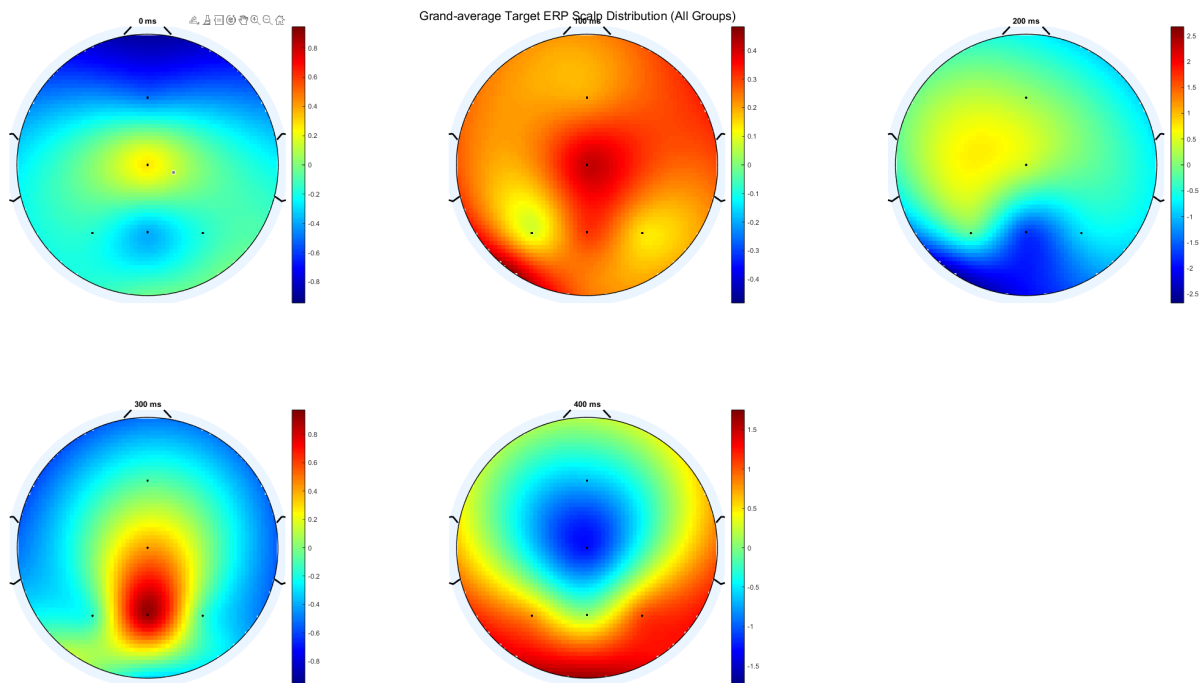
Part 2. Cross-Subject Analysis

ERP Response to Target and Non-Target Flashes

Using the complete set of training data from all groups, calculate and plot the grand average ERP, averaged over electrodes, for target and non-target flashes from 100 ms before to 600 ms after each flash. Mark the ERP components visible in the average ERP plots.



Plot the scalp map of the signal at 0, 100, 200, and 300 ms 400ms after the onset of the flash. As before, use topoplot.m and BCI.locs.



Compare these grand-average plots (average ERPs, topoplots) with your group's plots. Do these plots differ? If so, how do they differ and why do you think they differ in that way? If not, what makes them so consistent?

When looking at these new plots that incorporate all of the other groups' data, I have to conclude that the grand-average plots differ from our group's results with respect to both waveform clarity and spatial distribution, primarily due to the averaging effects and the intersubject variability. In the grand-average ERP, it definitely appears that the waveforms are noticeably smoother, with less high-frequency noise and overall a clearer component structure. I would guess that this smoothing occurs due to the averaging that is taking place across all of the participants, meaning that the sessions tend to cancel out idiosyncratic noise while also managing to preserve features that are consistent across subjects (such as what we see with the characteristic negative deflection near 300 ms and the prominent positive P300 that we see around 350 ms). We can also observe the presence of an additional negative deflection taking place around 600 ms that likely reflects later cognitive or motor processes that become more apparent when the signal-to-noise ratio improves through pooling.

As far as we are spatially concerned, I observed that the topoplots show differences as well. The grand-average data reveal a strong, well-defined parietal–occipital hotspot at 300 ms that was less distinct in the recordings for our group. I would wager that this improvement likely reflects

better statistical stability. Since we are aggregating multiple subjects, variability in electrode impedance, the head shape, and we are averaging out the attention, I bet this ends up emphasizing consistent neural sources that end up contributing to the P300. Furthermore, the presence of early activity near Cz near 100 ms would suggest to me the presence of a reliable sensory response across all the groups, whereas we can see that in our own earlier activity, this Cz occurred closer to a time of 0 ms, which would be consistent with our participants' slightly delayed responses. It seems to me like the overall delay and weaker parietal activation in our data can be explained by some of the individual differences in P300 latency and amplitude, which could include factors such as attention, fatigue.

Training and Testing Cross-Subject Decoding Models

Train a new model using both sessions of training data from all groups, except your own. This is referred to as the cross-subject model. Select a time range that is captured by each trial in your data matrix. Did you use the same time range as before? Why or why not?

Test the cross-subject model on your group's test data (test1 and test2) and on the test data from all other groups. How does the cross-subject model perform on a held-out subject (your group) compared to test data from subjects included in the training set? Also test your within-subject model on the test data from all other groups. How does the within-subject model perform on many held-out subjects compared to within-subject test data?

Accuracies	Group2 Test (Test Set 1+2)	All Other Groups Test
Cross-Subject Model	0.7278	0.7736
Within-Subject Model (Model1+2)	(Done in Part 1)	0.5802

I thought that it would be for the best to use the same time range of 100ms to 600ms for each of the trials. I thought to chose this window due to the other groups' data did not exhibit the delayed P300 latency that we observed in our participant. Since the canonical P300 component typically peaks somewhere between 300 and 400 ms, I believe that the standard 100 to 600 ms window would manage to capture both of the early sensory components (N1, P2) and the full P300 response for most subjects, making it more appropriate for such cross-subject training.

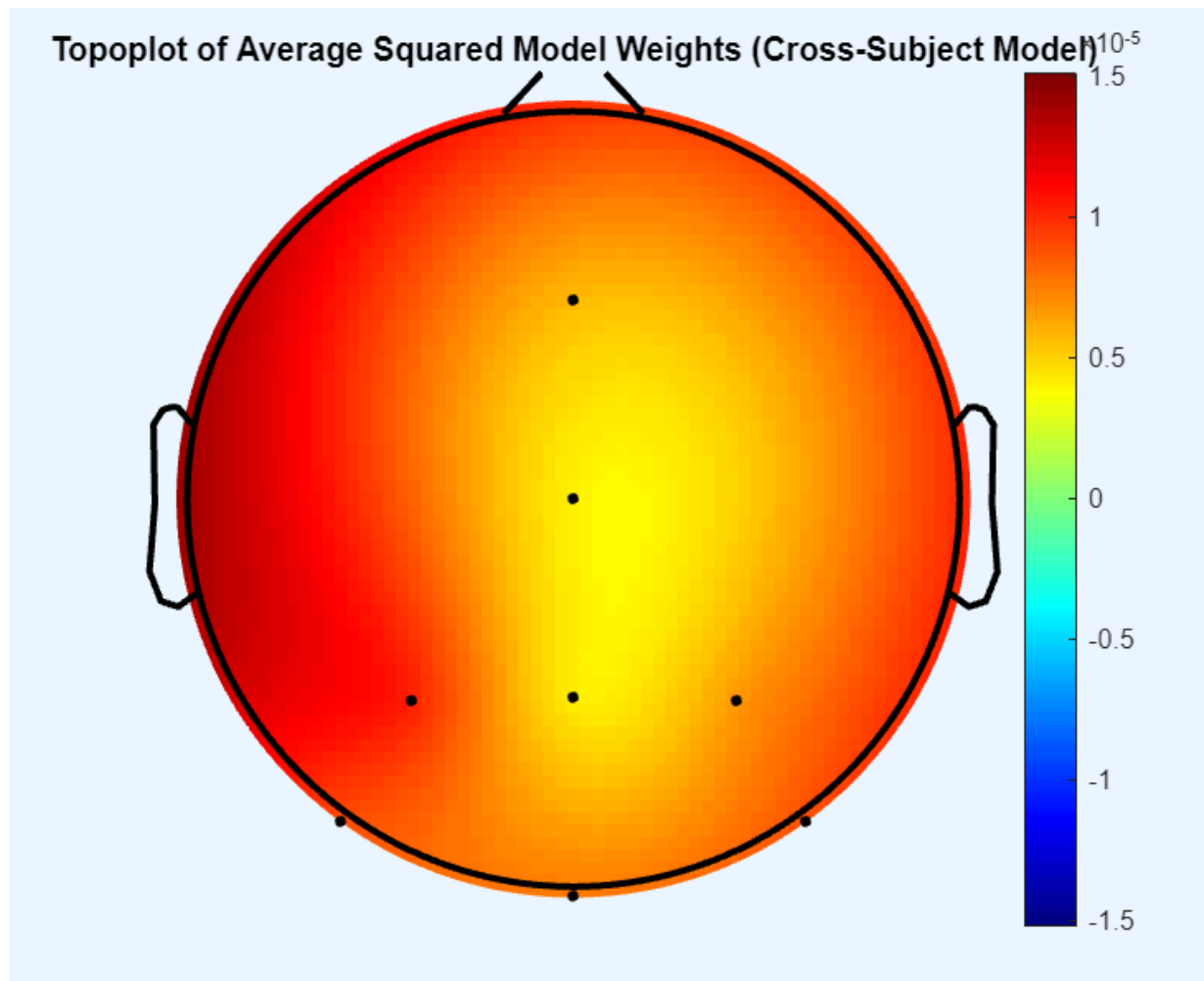
In terms of accuracy, the cross-subject model ended up achieving an accuracy of 0.7278 when the model was tested on our group's held-out data and 0.7736 on test data from the subjects that were included in its training set. I think that this would indicate that while the model managed to generalize reasonably well to a new subject, it honestly performed slightly better on participants

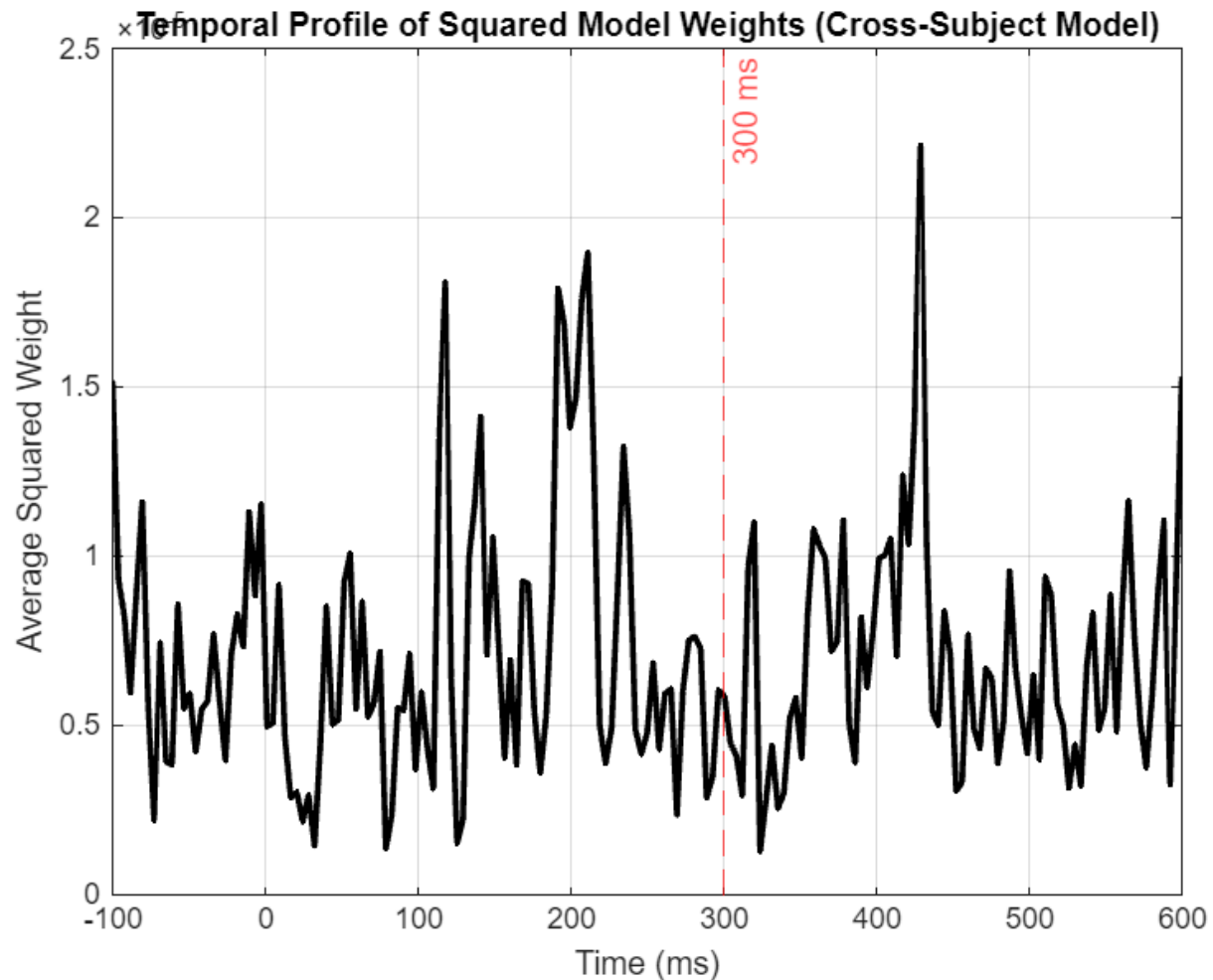
whose data distribution it had already encountered. I believe that the small performance gap should be expected, since variability across individuals (think of differences in P300 latency, amplitude, and the electrode impedance) would introduce some combination of subject-specific patterns that would end up limiting perfect generalization.

When it comes to the within-subject model, when it was tested on data from the other groups, it achieved an accuracy of 0.5802, which is considerably lower than its within-subject performance. It seems that this drop demonstrates that the model actually overfits to such subject-specific neural and noise characteristics, making it less capable overall of actually decoding these responses from the unseen participants. In contrast, the cross-subject model, which was trained on a diverse multi-subject dataset, learns features that are more generalizable across these individuals, which yields stronger performance on new subjects overall.

Analyze Cross-Subject Decoding Models

Use the cross-subject model weights to create a topoplot of the average squared weights at each electrode and a plot of the average squared weights over time. How do these plots of your cross-subject model compare to the equivalent plots of your within-subject model?





I would say that the topoplots of my cross-subject model actually looked very similar to those from my within-subject model, but I noticed that my own model showed noticeably stronger activity at the occipital electrode (Oz). I think that this could be due to the fact that my within-subject model was trained specifically on the data from my group, so it might have picked up on some individual characteristics of the recordings (for example, a slightly stronger occipital P300 response or different small differences in some of the electrode placements). It appears that the cross-subject model, on the other hand, was trained on many participants and therefore averaged out some of these individual differences, which ended up producing a more balanced spatial pattern that also still emphasizes these same posterior regions but less intensely at any one site.

When it comes to the temporal profile of the squared model weights, it seems as if my model had a larger average squared weight right at 300 ms, while the cross-subject model showed a broader response that included clear peaks at 200 ms and 400 ms. My within-subject model also had additional late peaks around 500 ms and 600 ms that just weren't as strong in the cross-subject

version. This makes sense to me, as we have already discussed that the P300 response in our group's data tends to be slightly delayed and more prolonged, while the cross-subject model was trained on people with more typical P300 timing. In short, the similarities show that both models seemed to capture the same sort of general P300 pattern, whereas the differences seem to reflect how the subject-specific personal neural timing and amplitude characteristics differ from the group average.

Discussion

Explain the advantages and disadvantages of using a within-session vs within-subject vs cross-subject model for a real-world BCI application.

In terms of the within-session model, it is trained and tested on the data that comes from the same recording session. Naturally, its main advantage is that it achieves the highest accuracy because it is going to always optimized for the exact conditions that took place during that particular session. Some examples may include: the same electrode positions, the noise characteristics, and the overall state of the participant. However, I would argue that this also makes it fragile, since even the small changes in the electrode impedance, or the user fatigue, or some environmental noise can likely drastically reduce the performance of the model. In such cases, it seems as if the model would need to be recalibrated frequently.

When it comes to a within-subject model that is trained on multiple sessions from the same person, it offers a sort of balance between accuracy and generalization. To this end, it can handle some of these aforementioned moderate variations in recording setup and user state while still managing to capture the wearer's unique neural patterns. In broad terms, this model seems like it would be well-suited for some of these personal BCIs in which the same user is going to be regularly interacting with the system. With that said, however, the main disadvantage is the amount of training data that would end up being required from that individual before reliable performance is going to be achieved.

The last model, a cross-subject model, is trained on the data that comes from many users and is then applied to a new person. In terms of its strengths, it is definitely scalable, since it enables rapid setup with little effort necessary for calibration, which would, of course, be ideal for any sort of plug-and-play BCIs or for any system that was using large deployments. However, in terms of the model's performance, it usually ends up being lower on the unseen subjects due to the nature of the inter-individual variability, such as the brain anatomy and the signal strength. In short, within-session models are going to be more accurate but ultimately more fragile, within-subject models are pretty reliable but are going to be more data-hungry, and cross-subject models are generalizable but tend to be less precise overall.